PRD: Image Change Detection Marketing Website

Product: [BRAND NAME], detecting changes between images captured at different times **Document type:** Product Requirements Document (website v1.0) **Status:** Draft for review **Owner:** [Name] **Last updated:** November 6, 2026

1. Overview

[BRAND NAME] compares images of the same place or object taken on different dates and highlights what changed: new construction, land-use shifts, deforestation, flooding, crop growth, infrastructure damage, site progress. It outputs clear, measurable, exportable change maps.

This PRD covers the **marketing and lead-generation website** for the product. The site must explain the value in seconds, let visitors experience the core idea through interaction, and convert qualified visitors into demo requests or trial users.

2. Problem Statement

Teams that rely on imagery over time (planners, insurers, agritech, forestry, construction, environmental agencies) compare images manually. This is slow, subjective, and error-prone. Lighting, camera angle, and seasonal variation create false alarms, while small but important changes get missed.

The website has to make this pain obvious and show that [BRAND NAME] removes it, without hype or jargon.

3. Goals and Non-Goals

Goals

1. Communicate the product's value within the first 5 seconds on screen.
2. Convert visitors into qualified leads through one primary action: **Request a demo / Try with your images**.
3. Build credibility through precise, calm, data-led design and copy.
4. Deliver a premium, agency-grade experience on mobile first, then desktop.

5. Meet Lighthouse 90+ across Performance, Accessibility, Best Practices, and SEO.

Non-Goals (v1.0)

- Full self-serve product app, dashboard, or billing.
- Multi-language support (English only at launch).
- Blog or CMS-driven content (planned for v1.2).
- Live image processing on the site (the interactive demo uses pre-built sample scenes).

4. Target Users and Personas

Persona	Role	Core need	What convinces them
Urban Planner	Municipal / consultancy	Detect unauthorized or unplanned construction	Accuracy, area measurements, audit trail
Agri / Forestry Analyst	Agritech, NGO, government	Track vegetation, deforestation, crop health	Seasonal-noise filtering, batch scale
Insurance Assessor	Claims and risk	Verify damage before and after events	Reports, evidence-grade exports
Infrastructure / Construction Manager	Site operations	Monitor progress and deviations	Drone and site photo support, speed
Researcher / Data Scientist	Academia, R&D	Reproducible change analysis	API, GeoTIFF/CSV export, methodology

Primary buyer: Decision-maker at a team that already works with satellite, aerial, or drone imagery. **Secondary:** Technical evaluator who needs API and format details.

5. Value Proposition and Messaging

- **Positioning:** Time made visible. Every image is a moment; the product shows what moved between them.
- **Headline:** “See exactly what changed between then and now.”

- **Tone:** Precise, calm, confident. No hype words such as “revolutionary” or “magic.”
- **Proof points:** Alignment of mismatched images, filtering of shadows and seasonal noise, quantified results (area, percentage, class), exportable reports.

6. Success Metrics

Metric	Target (first 90 days)	Measurement
Visitor-to-lead conversion	3% or higher	Form submissions / unique visitors
Hero interaction rate	50% or higher	Slider drag or tap events
Timeline demo engagement	35% or higher	Scrubber interaction events
Average scroll depth	65% or higher	Analytics
Mobile bounce rate	Under 50%	Analytics
Lighthouse scores	90+ in all four	CI audit
Largest Contentful Paint	Under 2.5s on 4G	Field data
Qualified lead rate	40% or higher of submissions	Sales review

7. Scope and Functional Requirements

7.1 Site map (single page, anchor navigation)

1. Navigation
2. Hero with before/after slider
3. Problem statement
4. How it works (sticky storytelling)
5. Interactive timeline demo
6. Use cases
7. Outputs and report preview
8. Comparison (manual vs [BRAND NAME])

9. Social proof
10. FAQ
11. Final CTA with lead form
12. Footer

7.2 Requirements by section

FR-1 Navigation

- Sticky, minimal; wordmark left, links (How it works, Use cases, Output, FAQ), solid CTA right.
- Hides on scroll down, reappears on scroll up.
- Mobile: full-screen menu with staggered reveal; focus-trapped and closable with Escape.

FR-2 Hero

- Eyebrow, H1, subheading, primary CTA, secondary text link.
- Draggable before/after comparison slider with labeled dates (T1, T2). Auto-plays once on load, then is user-controlled by mouse, touch, and keyboard (arrow keys).
- Changed regions outlined in the accent color with a soft pulse.
- Subtle 3D depth: pointer-based tilt and layered parallax on desktop; scroll or orientation-based on mobile; CSS fallback on low-power devices.
- Trust row (logos or usage stat), replaceable with real data.

FR-3 Problem section

- Three numbered statements revealed line by line on scroll.

FR-4 How it works

- Four steps: Upload, Align, Detect, Report.
- Desktop: pinned left column, changing right visual. Mobile: stacked steps with the same animations.
- Each step has a matching visual state.

FR-5 Interactive timeline demo

- Horizontal scrubber with 4 to 5 dates across a single scene.
- Changed areas glow in the accent color; a live counter updates (for example, built-up area gained, vegetation lost).

- Touch drag and keyboard support. Pre-rendered scenes, no backend.

FR-6 Use cases

- 4 to 6 cards: urban growth, deforestation, disaster damage, crop health, infrastructure progress, insurance verification.
- Each card: tag, one-line outcome, mini before/after reveal on hover (desktop) or tap (mobile).
- Horizontal scroll on mobile.

FR-7 Outputs

- Mock report card: change map, percent area changed, class breakdown, confidence score, export options (PDF, GeoTIFF, CSV, API).
- Count-up numbers on viewport entry.

FR-8 Comparison

- Two-column table: manual review vs [BRAND NAME] across speed, consistency, false alarms, scalability, audit trail.

FR-9 Social proof

- 2 to 3 testimonials with name, role, organization; slow fade carousel with pause control. Placeholders flagged for replacement.

FR-10 FAQ

- Accordion with smooth height animation; one item open at a time; full ARIA support.
- Topics: supported image types, alignment needs, accuracy, API, data privacy, pricing.

FR-11 Lead form (final CTA)

- Fields: name, work email, organization, use case (select), optional message.
- Inline validation, clear error text, animated success state, spam protection (honeypot plus rate limit or CAPTCHA alternative).
- Submission goes to [CRM / email / form endpoint] and triggers a confirmation email.
- Trust microcopy: no credit card, reply within 1 business day, privacy link.

FR-12 Footer

- Wordmark, short description, links, contact email, social links, privacy and terms.

7.3 CTA placement

Primary CTA appears in the navigation, hero, after How it works, after Outputs, in a mobile sticky bar (after hero), and in the final section. The label stays identical everywhere.

8. Design Requirements

- **Palette (4 colors):** Bone #F4F1EA, Ink #0F1114, Stone #8B8A84, Signal #FF5B2E. Signal is reserved for "what changed," CTAs, and hover states.
- **Typography:** Serif display (Instrument Serif or Fraunces), Inter or Geist for UI and body, JetBrains Mono for labels and data. Fluid scale with clamp().
- **Layout:** 12-column grid, asymmetric compositions, hairline dividers, generous whitespace; spacing scale 4/8/16/24/40/64/96/160.
- **Shape and depth:** 4 to 8px radius, near-zero shadows.
- **Imagery:** Code-generated visuals only (SVG or canvas); no stock photos.
- **Motion:** GSAP + ScrollTrigger, Lenis smooth scroll, Three.js or CSS 3D for hero depth. Easing cubic-bezier(.22,1,.36,1); durations 0.6 to 1.2s. No bounce or flashing.
- **Desktop extras:** Custom cursor and magnetic CTAs (desktop only).
- **Anti-template rules:** No generic icon-grid rows, no purple/blue gradients, no centered-everything layouts, no filler copy.

9. Non-Functional Requirements

Performance

- LCP under 2.5s, CLS under 0.1, INP under 200ms on mid-range mobile.
- Total initial JS under 200 KB gzipped (excluding lazy-loaded 3D); lazy-load below-the-fold assets; font-display: swap.
- Page weight under 16 MB is not a concern; target under 1.5 MB initial load.

Accessibility (WCAG 2.2 AA)

- Semantic HTML, one H1, logical headings.
- Keyboard operable slider, scrubber, accordion, and menu; visible focus states.
- `prefers-reduced-motion` disables parallax, smooth scroll, and tilt; simple fades remain.

- Color contrast AA; Signal never the only carrier of meaning (pair with labels or outlines).

Responsiveness

- Designed at 375px first; breakpoints at 768, 1024, 1440+.
- Tap targets at least 44px; safe-area insets respected; no horizontal overflow.

SEO

- Title, meta description, Open Graph, Twitter cards, canonical URL.
- JSON-LD: Organization and SoftwareApplication.
- Sitemap, robots.txt, descriptive alt text.

Privacy and Security

- HTTPS only, secure headers (CSP, HSTS).
- Cookie and analytics consent where required (GDPR, India DPDP Act).
- Form data stored only in the designated CRM; privacy policy linked.

Browser support

- Latest two versions of Chrome, Safari, Firefox, Edge; iOS Safari 16+; Android Chrome.

10. Analytics and Tracking

Event	Trigger
`hero_slider_interact`	First drag, tap, or key use
`timeline_scrub`	First scrubber interaction
`cta_click`	Any CTA, with location property
`usecase_card_open`	Card reveal, with card name
`faq_open`	Accordion item opened
`form_start`	First field focus
`form_submit`	Successful submission

Event	Trigger
`form_error`	Validation failure, with field
`scroll_depth`	25, 50, 75, 100%

Tooling: privacy-friendly analytics (for example Plausible or PostHog), plus UTM capture saved with each lead.

11. Technical Approach

- **Stack:** Next.js (or Astro) with Tailwind CSS; GSAP, Lenis; Three.js only for the hero.
- **Hosting:** Vercel or Netlify with CDN; image assets inlined or optimized as AVIF/WebP.
- **Form backend:** [Serverless function → CRM / email provider].
- **Content:** Copy and data held in a single config file for easy swaps.
- **Quality gates:** Lighthouse CI, axe accessibility checks, cross-device testing before release.

12. Content Requirements (inputs needed)

- Final brand name, logo, domain.
- Real proof points: usage numbers, accuracy claims (must be verifiable), logos, testimonials.
- Pricing model or “contact for pricing” decision.
- Sample imagery pairs for the hero and timeline (or approval to use generated scenes).
- Legal copy: privacy policy, terms, data-handling statement.
- CRM and email destination for leads.

13. Milestones and Timeline

Phase	Deliverable	Duration
1. Discovery	Content inputs, brand decisions, final copy	1 week
2. Design	Style tiles, mobile and desktop layouts, motion specs	1.5 weeks

Phase	Deliverable	Duration
3. Build	Responsive build, animations, form integration	2 weeks
4. QA	Performance, accessibility, device testing, fixes	1 week
5. Launch	Deploy, analytics verification, handoff docs	0.5 week

Total: about 6 weeks.

14. Risks and Mitigations

Risk	Impact	Mitigation
Heavy 3D and animation hurt mobile performance	High	Lightweight CSS fallback, device detection, performance budget
Unverifiable accuracy claims damage trust	High	Use only claims backed by tests; show methodology link
Placeholder testimonials or logos ship by mistake	Medium	Launch checklist flags every placeholder
Low form completion	Medium	Short form, clear trust cues, A/B test CTA wording
Motion causes discomfort	Medium	Reduced-motion support, calm easing, no flashing
Scope creep into a full app	Medium	Keep demo pre-rendered; defer upload feature to v1.1

15. Release Criteria (Definition of Done)

- All FR items implemented and verified on iOS Safari, Android Chrome, and desktop browsers.
- Lighthouse 90+ in all categories on mobile.
- Zero critical axe accessibility violations; keyboard-only walkthrough passes.
- Form tested end to end, with lead arriving in CRM and confirmation email sent.
- Analytics events firing correctly.

- All placeholder content replaced or explicitly approved.

16. Future Roadmap

- **v1.1:** Upload-your-own-images instant preview (free tier).
- **v1.2:** Blog, case studies, and resource library.
- **v1.3:** Pricing page, self-serve signup, and documentation site.
- **v2.0:** Localization and industry-specific landing pages.

17. Open Questions

1. What is the final brand name and domain?
2. Which industry is the lead vertical (this changes the hero scene and use-case order)?
3. Is pricing public, gated, or contact-sales only?
4. Which accuracy and scale claims can be stated publicly?
5. Which CRM and email tools should receive leads?
6. Is there a regulatory or data-residency requirement for uploaded imagery?